

Dougal Shakespeare, PhD

Marketing Science & Analytics

[LinkedIn](#), [Research](#), [Website](#)

Berlin-based Computer Scientist (PhD) specialized in Causal Inference, Experimentation and **Marketing Analytics**. Currently, developing my Bayesian approach to **Marketing Science** with **PyMC Labs**.

SKILLS: Causal Inference (DAGs, DID, A/B Testing) | Experimental Design | ML | Factor Analysis | Bayesian Inference | Marketing Analytics | Media Mix Modelling | LTV Forecasting | Funnel Analysis | Churn Modelling | EDA | Inferential Statistics
TOOLING: Python (NumPy, pandas, PyMC, PyMC-Marketing, CausalPy, PyTorch) | SQL (Snowflake, PostgreSQL) | Excel | BI & Dashboarding (Tableau, Power BI) | Git | Claude Code | Docker | AWS | Java | Bash

PROFESSIONAL EXPERIENCE:

Professional Sabbatical - Self-directed Researcher

Jan 2026 – Present (Berlin)

- Primarily focused on furthering my expertise in Marketing Science and Analytic through a 4 week bootcamp, [Bayesian Marketing Analytics](#) with the developers of the Bayesian open-source Python package, PyMC-Marketing. This course focuses on developing hands-on expertise in implementing Media Mix Modeling, Bass Diffusion Models, LTV forecasting, Causal Inference from observational data and budget optimization through Bayesian, Pythonic approaches.
- Since the defense of my PhD, I have been working on 2x quantitative research articles analysing anonymised Deezer listening logs in collaboration with researchers from the University of Tokyo and the Centre d'analyse et de mathématique sociales (CAMS).

Research Scientist - [ANR-RECORDS w/ Deezer](#), [Computational Social Science Lab](#)

Nov 2021- Dec 2025 (Berlin)

- Designed natural experiments and credible counter-factuals to capture the effect of Deezer's RecSys on engagement and diversity (DAG → DID + propensity score matching).
- Applied Survival Models to model Deezer users' intra-session retention and analyse the impact of algorithmic novelty on churn.
- Designed novel, semantic KPIs to track the diversity of users' product interactions using audio, mood and usage-based embeddings.
- Cleaned, transformed and analysed Deezer listening logs at scale (10M+ events) in Python and SQL for exploratory data analysis.
- Designed a novel offline RecSys evaluation protocol to measure item embedding space interpretability and presented the framework to industry stakeholders at the Complex Networks conference.
- Trained and evaluated node embeddings end-to-end (SDNE, node2vec, DeepWalk) on user-item co-occurrence networks.
- Presented insights to cross-functional stakeholders (R&D, Academia, Media) and published research in peer-reviewed, high-impact factor (Q1) journals and conferences including Nature's Scientific Reports and ISMIR.

Quantitative Market Research Consultant (contracted) - [Solsten](#)

Mar 2025 - Jun 2025 (Berlin)

- Client-facing consultancy work -- contacted by Solsten on a project basis for my quantitative specialization lacking in-house.
- Drove 3x large-scale quantitative research programs for EA & PLAION focused on better understanding competitor strategy, multi-channel adoption and brand health and awareness for AAA franchises.
- Partnered with external recruiters to field 2x multi-geo (NA, EU, Asia) large-scale (20K+ respondents) player research surveys.
- Developed and evaluated Bayesian models in PyMC to analyse geo and personality effects and delivered insights to senior stakeholders from Marketing as actionable recommendations for campaign development and channel-spend allocation.
- Delivered research reports on time to the client and delivered workshop sessions explaining the main insights and takehomes to non-technical stakeholders from Marketing & Product.
- Boosted the efficiency of open-ended response processing by 18x through implementing reusable, NLP-based automated topic modelling and sentiment analysis pipelines in Python.

Data Scientist - [CNRS](#), [ANR-RECORDS w/ Deezer](#)

Nov 2020- Nov 2021 (Berlin)

- Built and owned data pipelines ingesting raw listening logs from S3 into Snowflake and designed dashboards in Tableau to visualise changes in Deezer users' platform adoption rhythms across different user segments.
- Cleaned, normalized and clustered platform adoption time-series to generate a novel typology of Deezer users' listening rhythms and presented insights to R&D.
- Leveraged methods from information theory (PPMI, entropy) to analyse statistical variations between users' product adoptions (RecSys, playlists, search) and their listening rhythms.

EDUCATION:

PhD, Computer Science & Mathematics, CAMS @EHESS, (2021 - 2025)

Thesis: "*Multi-affordance Dynamics on Music Streaming Platforms. A Case Study on Deezer*".

Supervisor : Prof. Camille Roth, Reviewers: Prof. Emila Gomez (EU Commission), Prof. Christine Bauer (University of Salzburg)

Masters, Sound & Music Computing, Music Technology Group, University Pompeu Fabra, (2019 - 2020)

Select modules: Advanced ML, Research Methods, Music Information Retrieval, Signal Processing

Thesis: "*Exploring Artist Gender Bias in Music Streaming*" - research published at RecSys's ImpactRS workshop

BSc, Computer Science, University of Liverpool, 1st Class Honours (2016 - 2019)

Select modules: Scripting Languages, Social Network Analysis, Object oriented coding, Database Management

Thesis: "*Expanding Elaine Walker's Chaos Melody Theory to a Live Coded (Algorave) Context*"

AI & ML PROJECTS:

Reframing the Filter Bubble: Addressing growing concerns that music recommender systems constrain listening diversity, this project - the cornerstone of my PhD - leveraged the rich multi-affordance engagement signals within Deezer users' listening logs to dive deep into how listening diversity varies under different information representations (semantic vs discrete) & temporal scales (intra-session, inter-session, inter-affordance). This research was published in Nature's Scientific Reports & open access dataset & code ([link](#)) were released to facilitate reproducibility.

ANR-RECORDS: An academia-industry hybrid research project that brought together researchers from both the Social Sciences & Computer Science as well as the two industry R&D labs namely Deezer & Orange. Leveraging anonymised, granular listening histories supplied by Deezer, the project aimed to both understand the diversity consumed on music streaming platforms, explore the measurable effects of recommendations & understand the relation between declared & actual listening habits.

For-You-2: Ongoing open source research project that aims to extend Resident Advisor's "For You" beyond exploitation & towards exploration. The project is currently in the proof-of-concept phase. Having already scraped extensive event data & meta-data ([link to the scraper](#)), the subsequent project phase will be geared towards developing multi-modal event embeddings (image, text, co-occurrence data) connected by a 2T architecture.

Gender Bias in RecSys: This project aims to understand the degree to which music recommender systems can induce artist gender bias by simulating classical CF-based RecSys on two open source, Last-FM listening event datasets (LFM-1b, LFM-360K). All code from the project is released publicly including the data wrangler to enrich the datasets with gender metadata from MusicBrainz ([link](#)) as well as the code to reproduce the experiments ([link](#)). This research was also published in the proceedings of the Impact RS Workshop at RecSys-2020.

PEER-REVIEWED PUBLICATIONS:

"Reframing the Filter Bubble Through Diverse Scale Effects in Online Music Consumption", D.Shakespeare, C.Roth, V. Chareyron, Nature's Scientific Reports, 2025, DOI: 10.1038/s41598-024-75967-0, ([paper](#), [code](#))

"Exploring Artist Gender Bias in Music Recommender Systems", D.Shakespeare, L.Porcaro, E.Gomez, C.Castillo, ImpactRS @RecSys, 2020, DOI:10.48550/arXiv.2009.01715, ([paper](#), [code](#))

"Interpreting Node Embedding Distances Through n-order Proximity Neighbourhoods", D.Shakespeare, C.Roth, CompleNet, 2025, DOI:10.1007/978-3-031-57515-0_14, ([paper](#), [preprint arXiv](#))

"Tracing Affordance and Item Adoption on Music Streaming Platforms", D.Shakespeare, C.Roth, ISMIR proceedings, DOI:10.48550/arXiv.2109.03538, ([paper](#))